



Leakage Assessment

Uganda NFC Tausi
Internal Shell use Only

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Summary Land-Use Classification & Deforestation Leakage Assessment

- Land-Use Classification (2018 baseline) confirms that outgrower plots were predominantly cropland, fallow, and grass/shrub mosaics, with no forest-level biomass inside any enrolled plot prior to project start.
- Multi-dataset approach (NDVI + DynamicWorld + ESRI + AGB) provides a consistent, conservative identification of cropland and non-cropland inside all plots.
- Forest baseline (2017) derived from $AGB \geq 80$ t/ha, clipped to the 5 km leakage zone, defines 20,937 ha of pre-project forest outside the project area.
- This serves as the denominator for all leakage analysis.
- Annual forest loss (2017–2024) detected using Hansen lossyear, intersected with the 2017 baseline forest, and filtered to ≥ 0.5 ha patches.
- Observed loss patterns are consistent with regional deforestation dynamics and seem not linked to the outgrower activity areas.

Leakage Assessment

Goal is to provide spatial analysis for plot-level land-use history and to demonstrate that no agricultural production was displaced into forested areas (deforestation leakage check)

Only the 1,305 ha enrolled outgrower plots (202511104_Ug_OG_PD_(Final).shp) are analyzed.

Land-Use Classification (2015–2025) > VVB requires:

- Plot-level land-use classification (2015–2025)
 - Prove plots were non-forest before project start

Deforestation Leakage Analysis (2017–2025) > VVB requires:

- Deforestation leakage analysis (5 km buffer)
 - Prove no agricultural expansion moved into forests

Leakage Assessment deliverables

1. Land-Use Classification (2015–2025)

- 10-year historical classification per plot (active cropland, fallow, abandoned, degraded).
- Use Chloris AGB as primary dataset; Landsat 7/8 or Sentinel-2 as fallback.
- Detect time series patterns (2015–2024) based on NDVI, AGB, and crop cycles.
- Check whether grouping plots into cohorts is acceptable; still provide cohort-based time series.

2. Deforestation Leakage Analysis (5 km Buffer, 2017–2025)

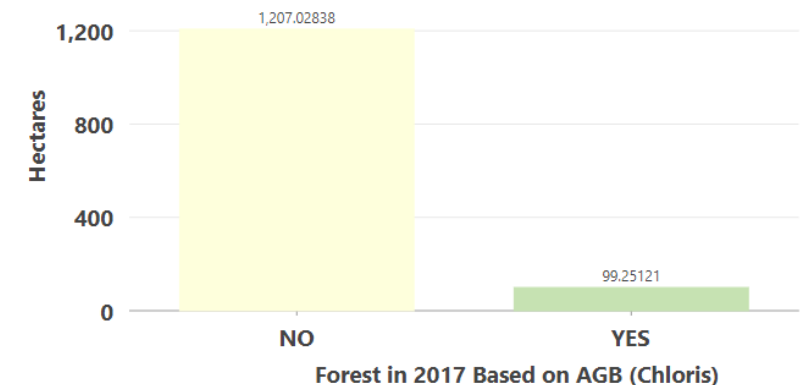
- Use Hansen tree cover loss (min patch >0.5 ha)
- Compare forest loss rate in 5 km buffer to district/regional baseline.

Preparation of outgrower data

Outgrower data and AGB 2017 overlay

1. 202511104_Ug_OG_PD_(Final).shp was used as final dataset to overlay the crop on.
 - This is the final output data of the previous outgrower eligibility study
 - However the eligibility rule of erasing overlap with AGB $\Rightarrow$ 250 t/ha for 2017 was not performed in previous analysis
 - This is done as well using the same methodology as extracting the 2007 AGB $\Rightarrow$ 250 t/ha
 - Classify AGB 2017 (Con ("Chloris\UG_Tausi_Stock_30m_2017_1.tif" $\geq$ 250, 1,0))
 - < 250 t/ha Non Forest (value=0)
 - $\Rightarrow 250$ /ha Forest (value=1)
 - Raster to Polygon, no simplification
 - Smooth Polygon, smooth tolerance 60m (~ 2 pixels)
 - Apply Buffer 5m, dissolve to merge adjacent shapes
 - Eliminate polygon parts $< 1200\text{m}^2$, remove holes
 - Identity to exclude forest polygons
2. An additional ~ 100 hectares of forest (AGB > 250 t/ha) in 2017 are within the outgrower
3. These are erased from them the outgrower areas for the lan-use classification
4. Final eligible outgrower areas used for crop analysis: UG_OG_PD_Clip_ID_AGB2017
 - Including a column FOREST_2017_AGB (YES/NO)

Summary of area Split by forest/Non Forest 2017



Land-Use Classification

Summary: Cropland (2018) in Outgrower Areas

Identify which parts of the outgrower areas were cropland in 2018 using a multi-dataset approach.

- Method (simplified):
 - Classify NDVI 2018: Class 3 = Primary Cropland Signal.
 - Validate structure with AGB 2018 (percentiles P25/P50/P75): cropland shows the lowest biomass
 - Cross-check with Dynamic World 2018 and ESRI LandCover 2018: both show the highest cropland proportion inside NDVI Class 3.
 - Build a consensus confidence map: cropland from NDVI, DW, ESRI; apply AGB > 80 t/ha woody mask (no cropland when woody)
- Why it's reliable:
 - NDVI cropland is ecologically consistent across years (2018–2025) in AGB statistics.
 - DW & ESRI provide independent agreement (though conservative in smallholder mosaics)
 - AGB acts as a structural check
- Result:
 - A confidence-weighted cropland map (2018)
 - Per-polygon reporting: Used Tabulate Area to calculate the % of each confidence class within every outgrower polygon
 - Temporal alignment (2018 for all inputs) is maintained

Optional??

- *Fine-tune woody threshold ? if local knowledge suggests.*
- *Repeat the same workflow for other years to quantify change.*

Land-use Classification

Due to cloud cover in Uganda between 2015–2018 and the small size of individual outgrower plots, cloud-free Sentinel-2 NDVI composites were not consistently obtainable.

Therefore, applied a multi-source land-cover consensus method using Dynamic World, ESRI LC and NDVI-based land cover classes. For all 2 datasets the year 2018 was used as a test start.

To identify which parts of the outgrower areas were cropland in 2018 a multi multi-dataset approach was used

Method (simplified):

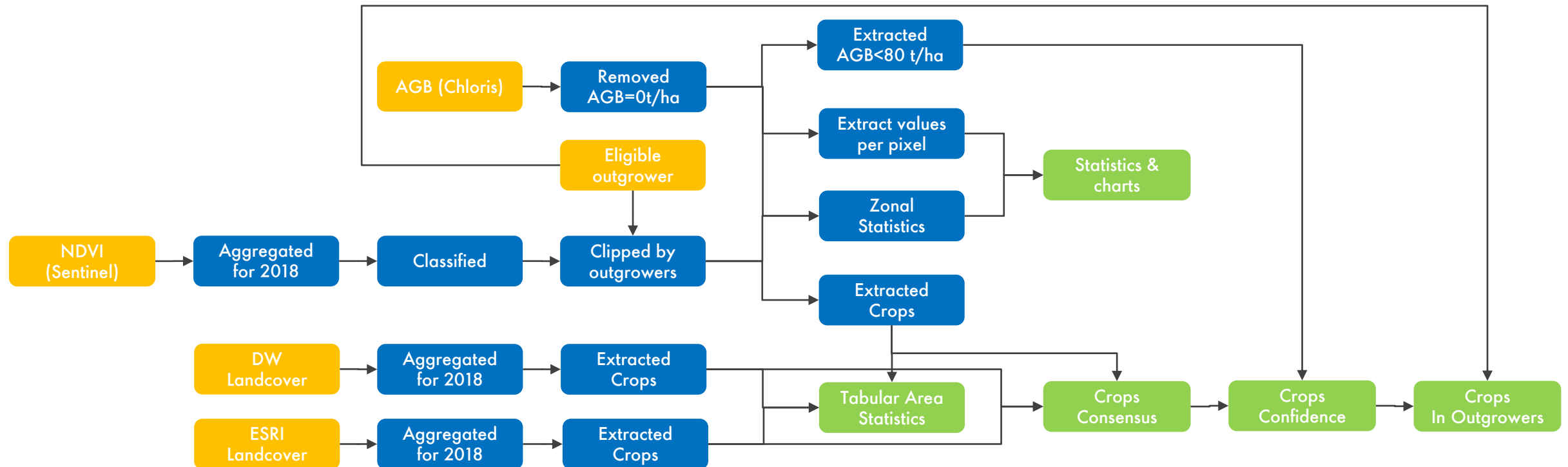
- Classify NDVI 2018: Class 3 = Primary Cropland Signal.
- Validate structure with AGB 2018 (percentiles P25/P50/P75): cropland shows the lowest biomass
- Cross-check with Dynamic World 2018 and ESRI LandCover 2018: both show the highest cropland proportion inside NDVI Class 3.
- Build a consensus confidence map: cropland from NDVI, DW, ESRI; apply AGB > 80 t/ha woody mask (no cropland when woody)

Land-use classification methodology

1. NDVI classification (Sentinel-2, 2018) was used as the primary cropland indicator, with NDVI class 3 representing Cropland
2. Above-Ground Biomass (AGB, 2018) was cleaned by masking non-vegetated pixels ($AGB \leq 0$ t/ha) and used to build a woody biomass mask ($AGB > 80$ t/ha).
3. NDVI class behaviour was validated using AGB percentiles (P25/P50/P75/P90), confirming that NDVI class 3 consistently shows the lowest biomass distribution across all years and therefore corresponds to cropland.
4. Dynamic World Landcover (2018) and ESRI Landcover (2018) were overlaid to provide independent vegetation/cropland context
5. Tabulate Area, boxplots, and percentile line charts were used to compare NDVI classes with AGB, ESRI, and DW, confirming internally consistent land-cover structure.
6. A consensus cropland confidence map was derived from NDVI, DW, ESRI, reclassified to cropland=1 & non-cropland=0 and a AGB woody mask, producing uncertainty classes (non-cropland, uncertain, medium confidence).
7. Cropland statistics inside the outgrower areas were extracted using the consensus layer

NDVI provides the primary cropland signal; DynamicWorld & ESRI provide supporting evidence; AGB provides a woody-structure

Land-use classification methodology



NDVI Classification

The 2018 NDVI image was selected as the baseline year because planting started in 2017 and 2018 is the first full season observable in Sentinel-2. NDVI (Normalized Difference Vegetation Index) is widely used to identify vegetation types based on canopy greenness and density.

The thresholds applied in this classification follow ecological patterns typical for East African savanna landscapes and mixed agroforestry systems. These systems commonly show strong contrasts between sparse vegetation, grass/shrub mosaics, active cropland, and woody vegetation.

Research on East African smallholder systems shows that croplands often fall in the NDVI range 0.35–0.50, depending on crop type, soil brightness and patch size. Sparse vegetation and shrub/grass mosaics typically fall below 0.35, while dense woody vegetation generally exceeds 0.50.

To reflect these patterns, the 2018 NDVI was reclassified into five land-cover categories:

NDVI Classification 2018			
Class	Threshold	Class	Explanation
0	< 0.05	Other	Bare soil, non-vegetated areas, water, or infrastructure
1	0.05-0.2	Degraded/Sparse	Sparse vegetation, degraded areas, early-season regrowth
2	0.2-0.35	Grassland/Shrubland	Grass, herbaceous vegetation, shrub mosaics, low woody density
3	0.35-0.5	Primary Cropland Signal	Active cropland and field mosaics mixed with small trees/shrubs
4	>0.5	Forest	Dense vegetation, forest, or areas with substantial woody biomass

NDVI Statistics (2018-2025)

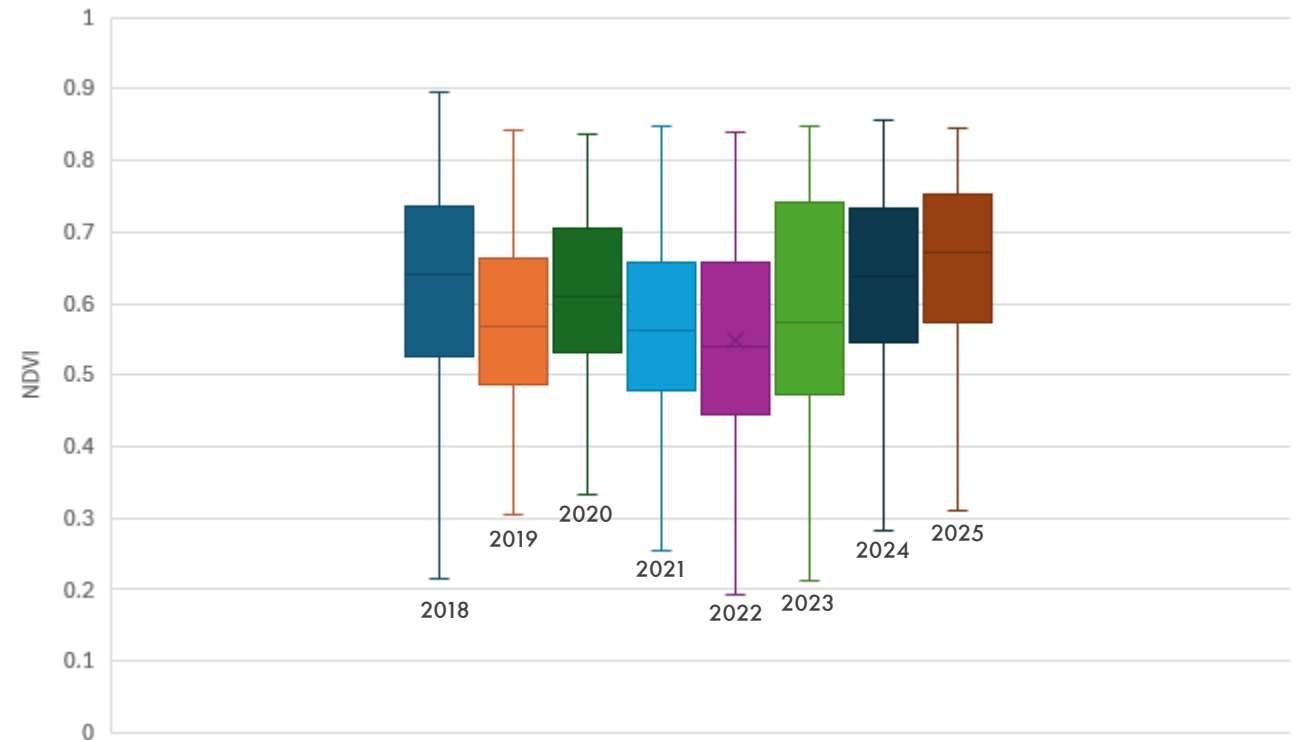
Interpretation

- NDVI distributions from 2018–2025 show stable low–medium vegetation activity typical of smallholder cropland and fallow land in Uganda.
- No forest-like NDVI (>0.7 sustained) appears in any year, consistent with pre-project non-forest conditions across the enrolled plots.

Methods

- Annual NDVI values are based on cloud-masked Sentinel-2 median composites (40% cloud threshold). Each box represents the distribution across all plots.

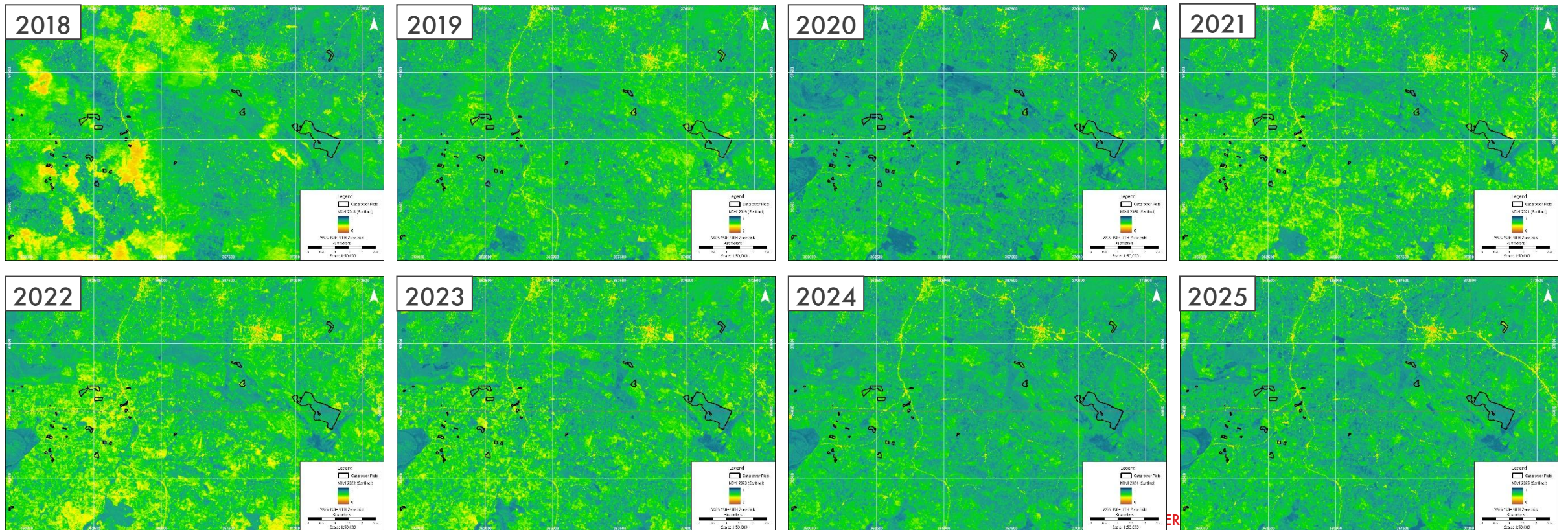
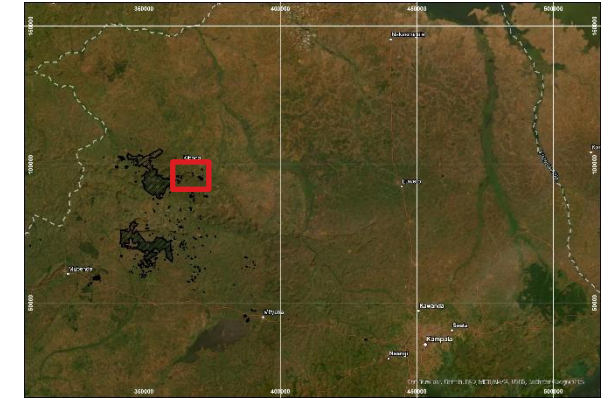
NDVI Trends Across Outgrower Plots (2018–2025)



NDVI (Sentinel) Maps Series

Annual median NDVI composites (cloud-masked Sentinel-2, 40% threshold) for 2018–2025, clipped to the project region. NDVI patterns remain stable through the period

- Higher NDVI values (green) indicate active vegetation or crop cover
- Lower values (yellow–brown) indicate bare soil, sparse vegetation, or fallow.

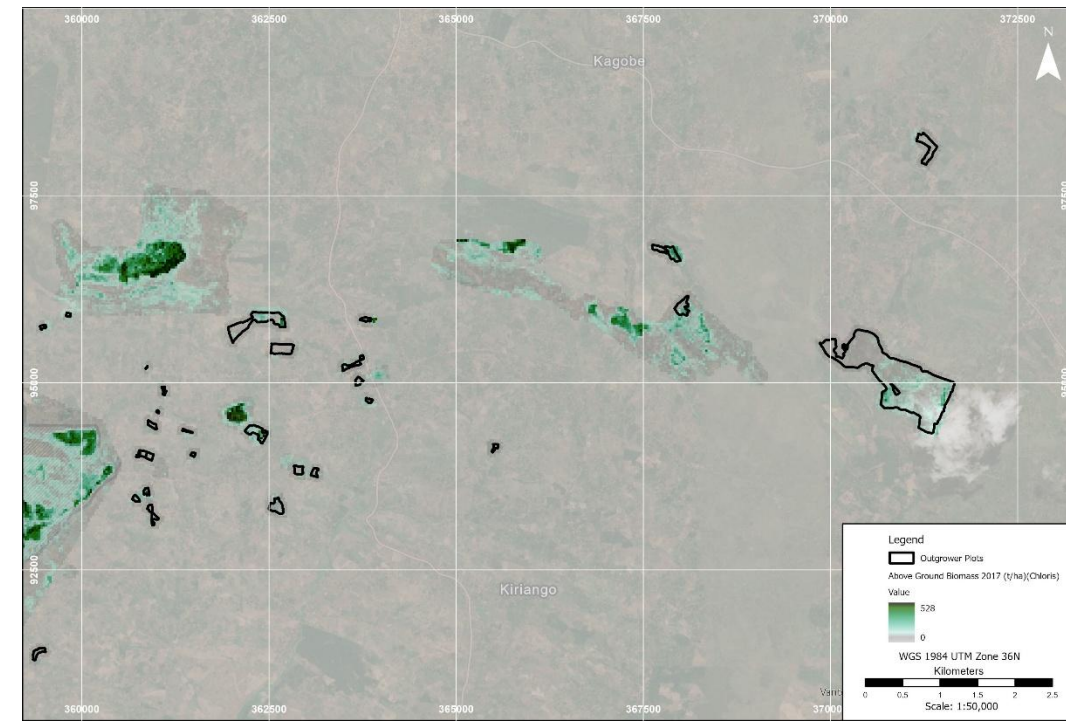


AGB Distribution (2017 – Pre-Project Condition)

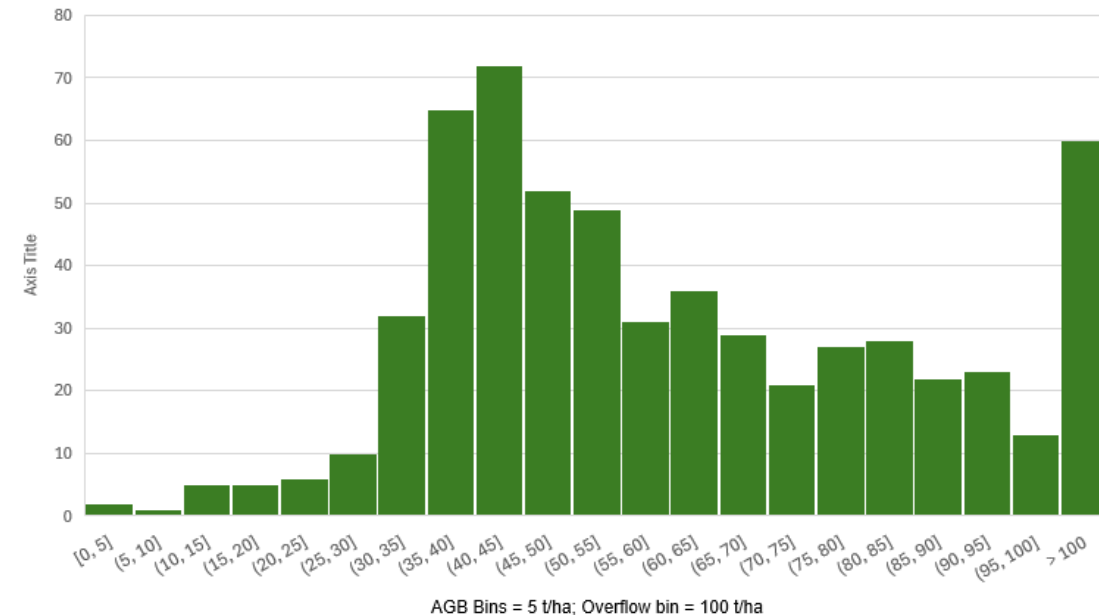
Interpretation: Above-ground biomass in 2017 is concentrated well below forest levels. The distribution reflects smallholder mosaics (cropland, fallow, shrub/hedge trees). Few values exceed the woody threshold (70 t/ha), confirming plots were non-forest immediately before project start.

The vertical line at 80 t/ha marks a conservative forest threshold for Uganda; values above this are rare within enrolled plots.

Woody pixels (AGB > 80 t/ha) are masked later for classification; histogram shows raw AGB 2017.



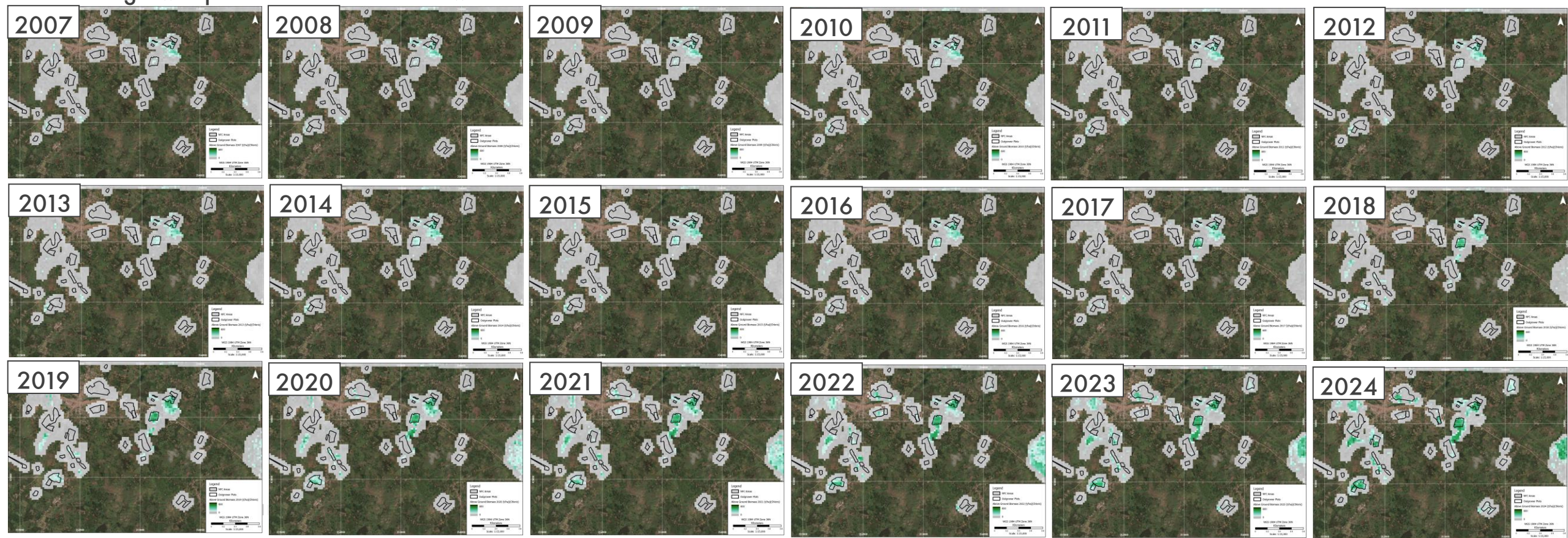
AGB Distribution (2017 - Pre-Project Condition)



Above Ground Biomass (Chloris) Maps Series

Annual Above-Ground Biomass (AGB) estimates from Chloris (10 m) for 2007–2024.

- Higher AGB values (darker greens) indicate denser woody vegetation
- Lower values (light greens/greys) represent cropland, fallow, or shrub/grass mosaics.
- Outgrower plot boundaries are shown for reference.



NDVI and Above Ground Biomass

Statistics of Above-Ground Biomass (AGB, Chloris 2018) were calculated per NDVI land-cover class to understand how vegetation relates to the NDVI-based classification.

- A box plot was used to visualize the full distribution of AGB values per NDVI class. This allows comparison of medians (P50), interquartile ranges (P25–P75) and extreme values/outliers. The box plot highlights how AGB increases from low-NDVI classes toward woody vegetation and confirms that NDVI Class 3 (Mixed/Cropland) consistently has the lowest biomass distribution.
- A percentile line plot (P25, P50, P75) was used to examine differences between NDVI classes. Percentiles provide a clearer picture of vegetation structure than means. P25/P50/P75 across classes shows a stable and separation between cropland, grass/shrub mosaics and woody vegetation.

Class	Class Description	Mean	STD	Median	PCT25	PCT50	PCT75	PCT90
0	Other	108	87	83	53	83	126	227
1	Degraded/Sparse	111	91	83	54	83	125	227
2	Grassland/Shrubland	118	100	80	53	80	135	280
3	Primary Cropland Signal	96	85	67	49	67	99	169
4	Forest	117	110	69	49	69	124	340

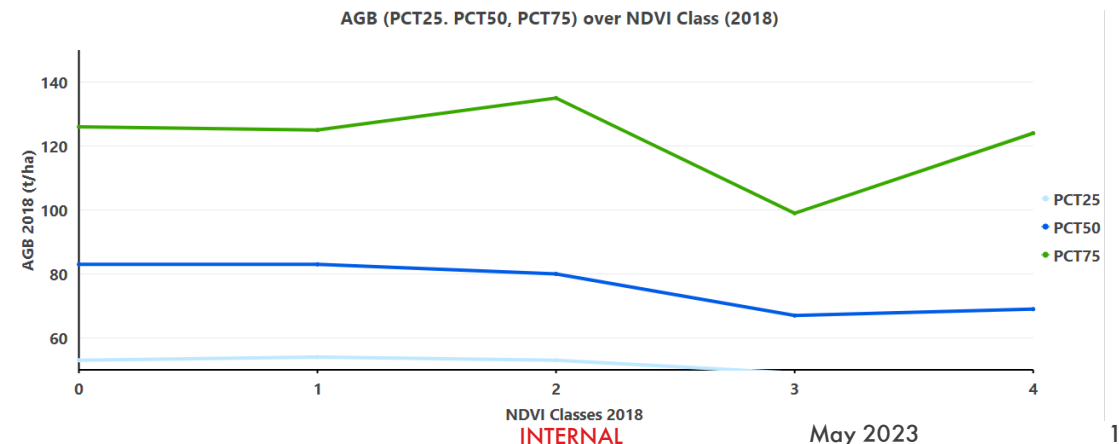
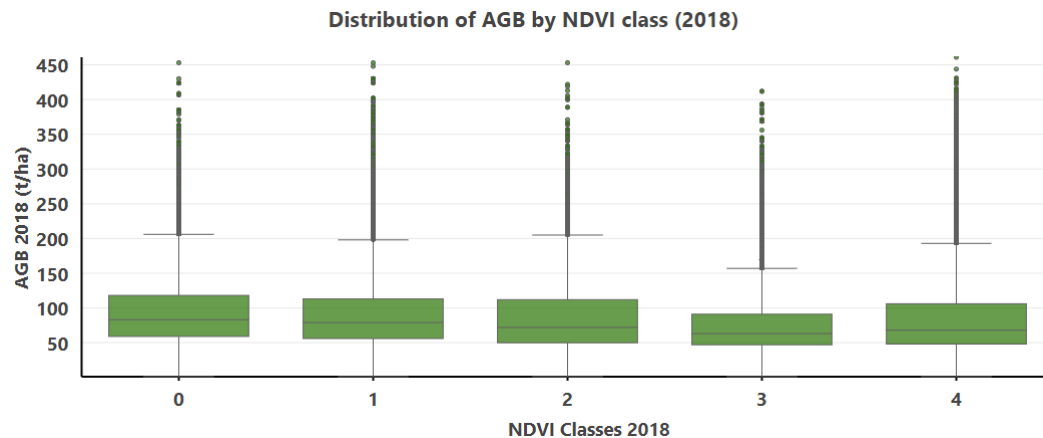
AGB values shown here are pixel-level; plot-level mean AGB used for forest checks remains below forest thresholds.

NDVI and Above Ground Biomass Relationship

The boxplots and percentile lines show a clear and consistent ordering of biomass across classes, confirming the ecological logic of the NDVI classification.

- NDVI class 3 (Primary Cropland Signal) has the lowest woody biomass at all percentiles.
- NDVI classes 2 and 4 show higher P75/P90 values, reflecting shrub and forest structure.
- Forest (class 4) has the heaviest upper tails, indicating high woody biomass.
- Degraded/Sparse sits between cropland and shrubland.
- Percentile curves confirm: Cropland < Degraded/Sparse < Grassland/Shrubland < Forest
- This validates the NDVI thresholds as reliable indicators of vegetation structure

Together, the box plot and percentile profiles demonstrate that NDVI Class 3 aligns with low-biomass cropland areas, while NDVI Classes 2 and 4 correspond to grass/shrub mosaics and woody vegetation



NDVI Cross-Checks with DynamicWorld & ESRI Landcover

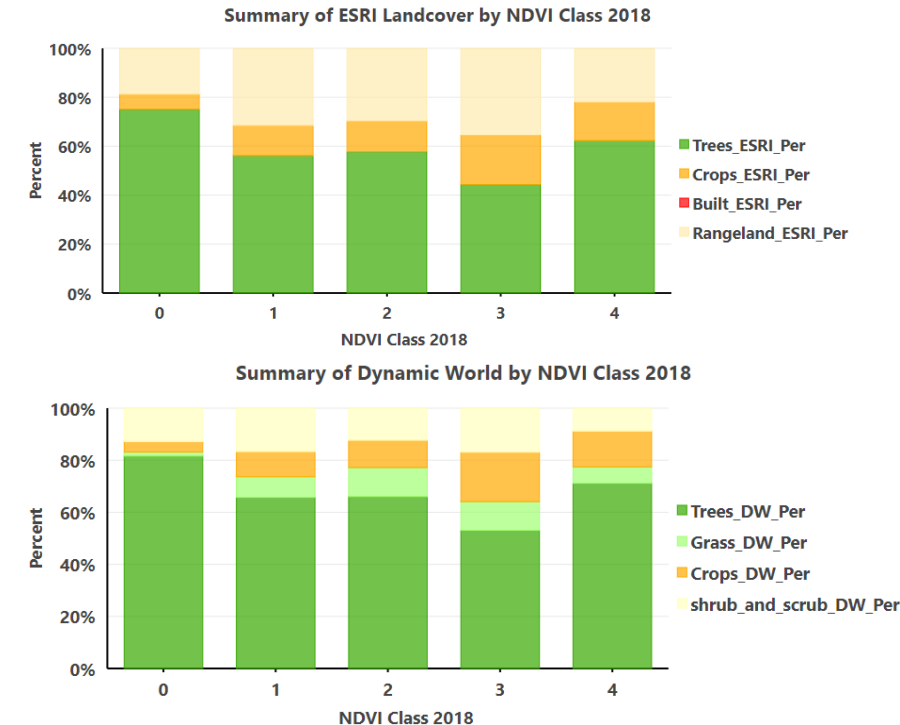
To independently verify whether the NDVI-based land-cover classes behave consistently with other global land-cover datasets.

DynamicWorld (DW) and ESRI LandCover were selected because they are Sentinel-2 based, available for 2018, and represent both vegetation structure (DW) and thematic land cover (ESRI).

Tabulate Area was performed to measure the percentage of each DW/ESRI class inside each NDVI class (Normalised percentages). These quantify how much each independent land-cover type is present inside each NDVI class. This provides a validation of NDVI behaviour.

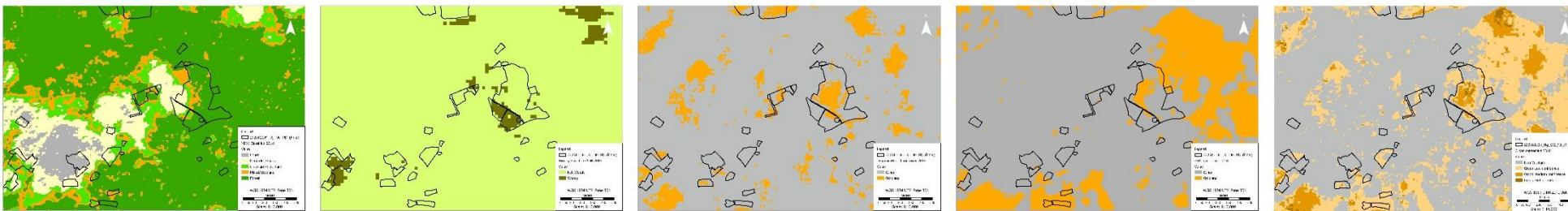
Conclusions

- ESRI Land Cover (2018)
 - NDVI class 3 has more ESRI cropland than any other NDVI class
 - All classes show a mixture of trees and rangeland
- DynamicWorld (2018)
 - NDVI class 3 shows the highest proportion of Crops
 - All classes show a mosaic mixtures of trees, grass and shrub
 - DynamicWorld tends to underdetect cropland in small fragmented fields (<20 m)
- Both DynamicWorld 2018 and ESRI LandCover 2018 show that NDVI class 3 contains the highest cropland signal
- NDVI classes 0,1,2,4 show lower cropland signals for both datasets
- The cross-checks support the validity of the NDVI-based cropland class for 2018



Integrated Interpretation

- NDVI (2018) Primary Cropland Signal
 - NDVI class 3 (0.35–0.50) shows the lowest AGB across percentiles (P25–P90)
 - This matches expected cropland vegetation structure in East African smallholders
- Above-Ground Biomass (AGB 2018) > Woody Structure Check
 - Forest and shrubland classes show much higher upper biomass tails
 - $\text{AGB} > 80 \text{ t/ha}$ reliably indicates woody areas inconsistent with cropland > woody filter
- DynamicWorld (2018) - Vegetation Form Cross-Check
 - NDVI class 3 contains the highest proportion of DW “Crops”, plus grass/shrub mosaics typical of smallholder fields
 - DW confirms the spectral/vegetation logic of NDVI classes
- ESRI LandCover (2018) - Thematic Cropland Cross-Check
 - ESRI cropland peaks in NDVI class 3 compared to classes 0–2



A combined consensus approach gives a cropland map for 2018.

Consensus Cropland Map (2018)

Conclusions from datasets:

- NDVI class 3 is the strongest and most validated cropland signal
- DynamicWorld 2018 provides additional spectral vegetation evidence
- ESRI LandCover 2018 adds thematic land-cover confidence
- $AGB > 80 \text{ t/ha}$ identifies woody contradictions inconsistent with cropland
- Combining these creates multi-dataset confidence score

Inputs (binary):

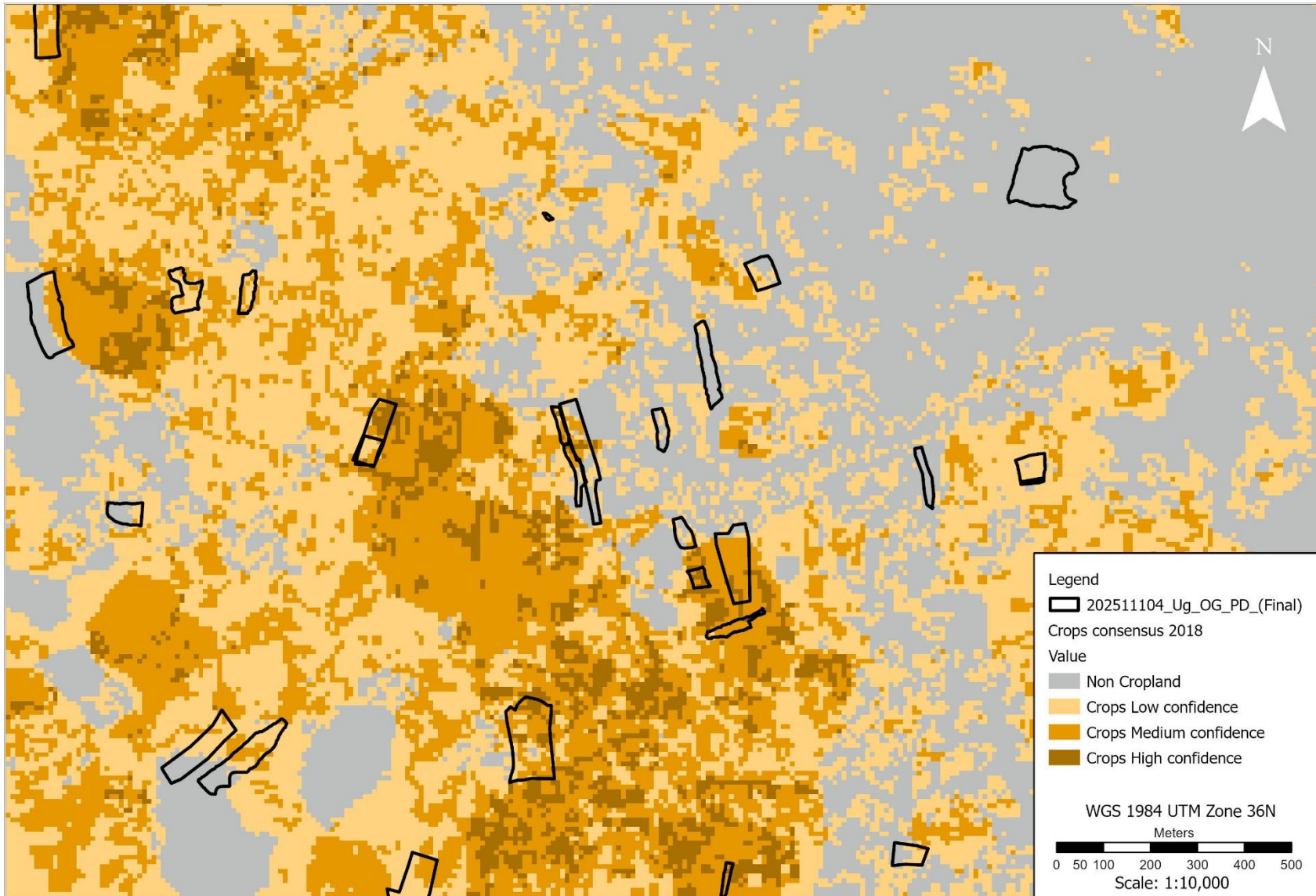
- $NDVI_{crop} = 1$ if NDVI class 3
- $DW_{crop} = 1$ if DW Crops
- $ESRI_{crop} = 1$ if ESRI Cropland
- $WOODY = 0$ if $AGB > 80 \text{ t/ha}$

Classes:

- 0 = Non-cropland or woody contradiction
- 1 = Uncertain cropland (NDVI only)
- 2 = Medium confidence (crops in 2 dataset)
- 3 = High confidence (NDVI + DW + ESRI)

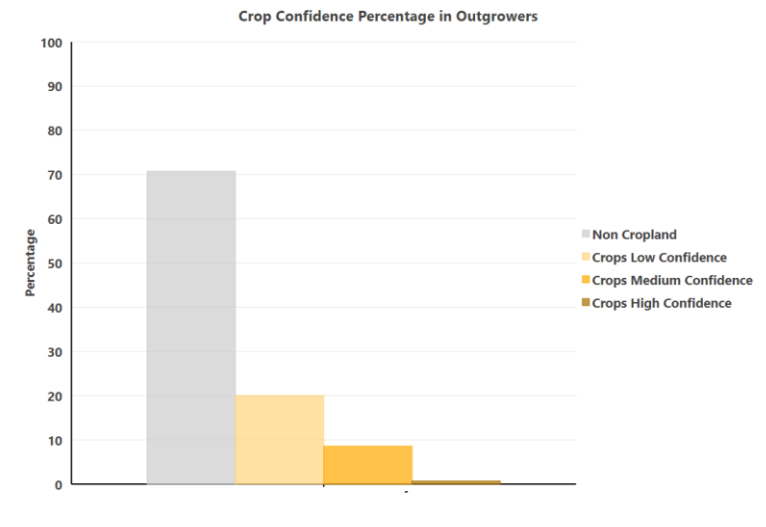
$$CONSENSUS = (NDVI_{crop} + DW_{crop} + ESRI_{crop}) * WOODY$$

Cropland Confidence Inside Outgrower Plots (2018)



Per-plot cropland confidence was computed by overlaying the consensus cropland raster with the outgrower polygons using Tabulate Area. This provides a per-polygon breakdown of High, Medium, Low confidence cropland and non-cropland/woody areas.

- High confidence = NDVI + ESRI + DW
- Medium confidence = 2 datasets
- Low confidence = NDVI-only cropland
- Non-cropland/woody = AGB > 80 t/ha or NDVI class < 3



Summary

The combined analysis of NDVI, Above-Ground Biomass (AGB), DynamicWorld, and ESRI LandCover provides cropland distribution in the outgrower areas for 2018. NDVI class 3 (0.35–0.50) shows the lowest AGB values across all percentiles, confirming that this class corresponds to low-woody, potentially used cropland. In contrast, shrubland and forest classes show higher upper percentiles, reflecting increasing woody biomass and confirming the distinction between cropland and woody vegetation.

Cross-checks with DynamicWorld (2018) and ESRI LandCover (2018) show the highest proportion of cropland-related classes within NDVI class 3. The relative pattern across datasets aligns with the NDVI-based classification.

AGB adds a filter: $AGB > 80 \text{ t/ha}$ reliably identifies woody areas incompatible with cropland.

Together, these datasets creates a multi-source interpretation:

- Cropland = NDVI class 3, supported by DW/ESRI where present, and structurally confirmed by low AGB values.
- Non-cropland = NDVI classes 0–2 OR any pixel with woody biomass ($AGB > 80 \text{ t/ha}$). This includes grassland, shrubland, fallow areas, paths, and other non-cropland mosaics.

To integrate the strengths of each dataset, consensus approach was applied, combining NDVI (primary signal), DW and ESRI (supporting evidence), and AGB (woody). This produces a cropland map for 2018, capturing both uncertainty and agreement across datasets while ensuring that only non-woody pixels are retained as cropland.

Data

Data input:

- 202511104_Ug_OG_PD_(Final).shp
- DymanicWorld Landcover 2018
- ESRI Landcover 2028
- Chloris Stock 30 m 2017
- Chloris Stock 30 m 2018
- NDVI Sentinel 2018

Data Ouput:

- UG_OG_PD_Erase_AGB2017_20260301.shp (202511104_Ug_OG_PD_(Final).shp removed forest in 2017)
- EXCELL

Deforestation Leakage Analysis

Deforestation Leakage Analysis (5 km Buffer, 2017–2024)

Assess whether agricultural activities from enrolled outgrowers caused displacement of land-use pressure into nearby forest areas.

To confirm no project-driven deforestation beyond the project boundary.

Establish a pre-project forest baseline (2017) and track annual forest loss relative to this baseline.

Compare forest-loss rates in the leakage zone with district-level trends to contextualise results.

Start: Entire landscape

⇓

Step 1: Remove everything that is not forest (AGB mask)

⇓

Step 2: Remove everything outside the 5 km buffer

⇓

Step 3: Remove the project plots

⇓

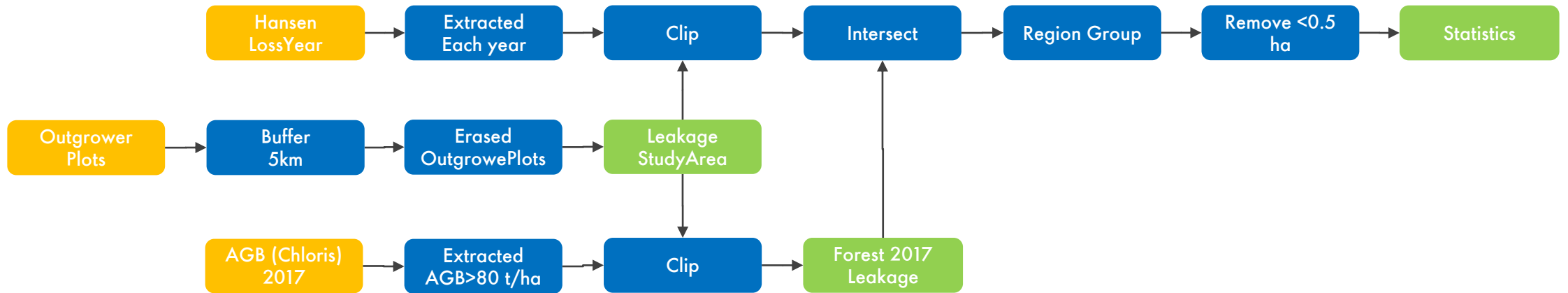
Result = Forest_2017_LZ

This is your initial forest baseline for leakage.

Deforestation Leakage methodology

1. Define Leakage Zone
 - 5 km buffer around all enrolled outgrower plots (dissolved)
 - Outgrower plots removed from the buffer to analyse forest outside the project area
2. Build 2017 Forest Baseline (Forest_2017_LZ)
 - Forest identified using $\text{AGB } 2017 \geq 80 \text{ t/ha}$ (Chloris AGB)
 - Forest_2017 clipped to the 5 km buffer minus the plots.
 - Baseline forest area (InitialForestArea) = 20,937 ha
3. Annual Forest Loss (Hansen “lossyear”)
 - For each year (2017–2024):
 - Extract pixels where $\text{lossyear} == y$.
 - Clip to leakage zone.
 - Intersect with Forest_2017_LZ (ensures only forest-derived loss is counted)
 - Apply a 0.5 ha minimum mapping unit (Region Group + COUNT ≥ 6).
 - Convert pixel count $\times 0.09 \text{ ha} \rightarrow$ annual forest loss (ha)
4. Compute Annual Loss Rate
 - $\text{LossRate}_{\%} = \text{CleanArea}_{\text{ha}} / \text{InitialForestArea} \times 100$
5. Optional Baseline Comparison
 - Compute district-level loss rate using same forest definition + MMU filter.

Deforestation Leakage methodology



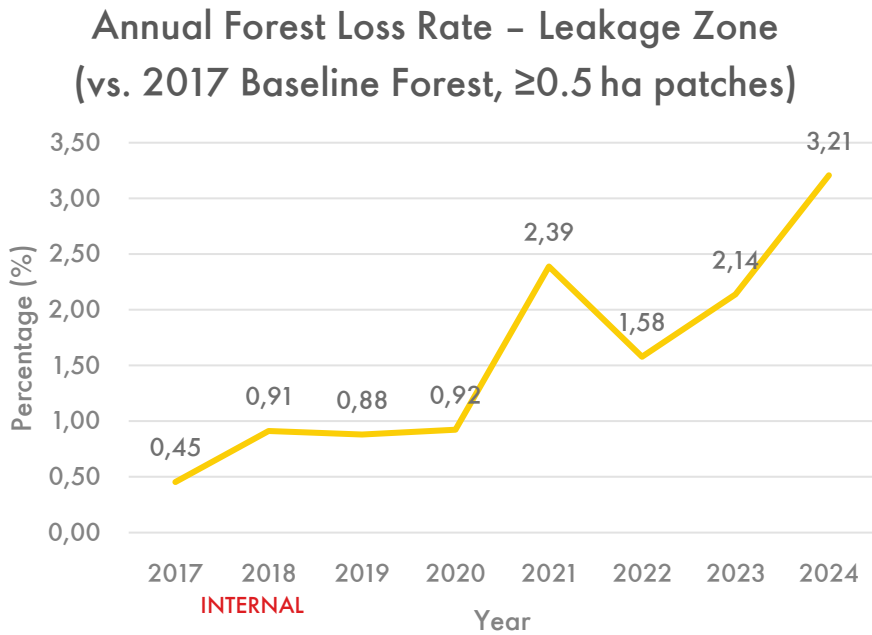
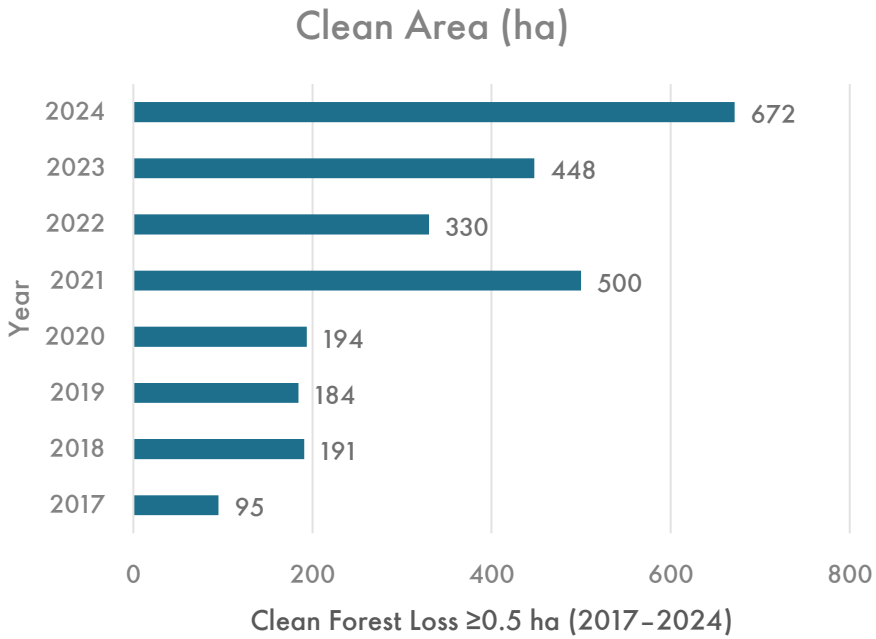
Forest Loss 2017–2024 (Leakage Zone)

Clean forest loss (≥ 0.5 ha patches) within the 5 km leakage buffer shows modest year-to-year variation, with most years below 500 ha of loss. 2024 shows the highest loss (~672 ha), but this remains less than 3.3% of the initial 2017 forest baseline (20,937 ha).

Observations:

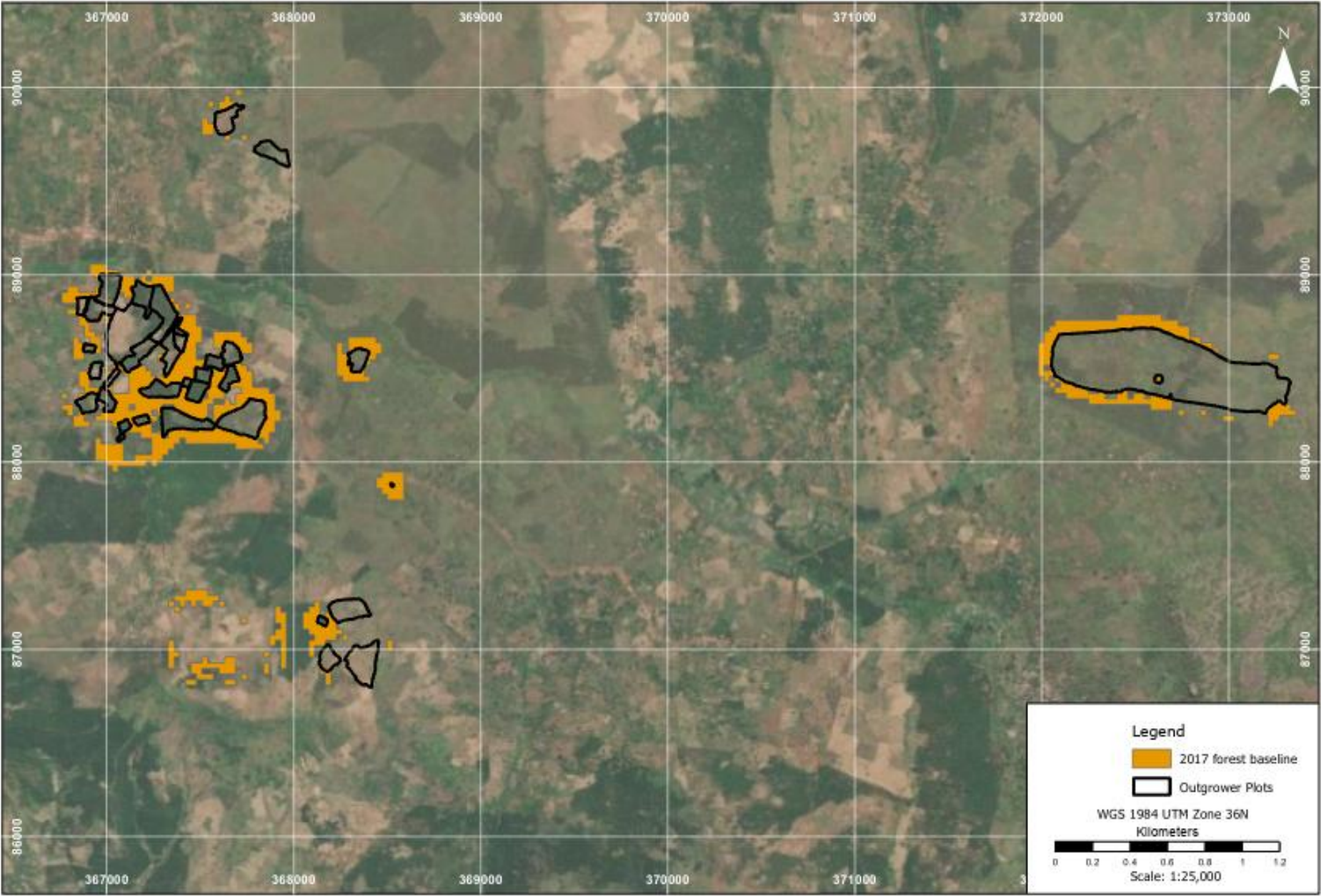
- Loss rates in 2017–2023 remain <3%
- 2024 shows a slight increase to 3.21%
- MMU filtering removes 12–34% of Hansen noise

Year	Initial Forest Area (ha)	Raw Area (ha)	Clean Area (ha)	Loss Rate (%)	% Removed (MMU)
2017	20937	144	95	0.45	34.08
2018	20937	242	191	0.91	21.26
2019	20937	223	184	0.88	17.46
2020	20937	251	194	0.92	22.88
2021	20937	570	500	2.39	12.25
2022	20937	411	330	1.58	19.65
2023	20937	516	448	2.14	13.17
2024	20937	752	672	3.21	10.75



Forest baseline

2017 forest baseline
inside the 5 km leakage
zone (AGB $\geq$ 80 t/ha).



Hansen maps

Map 1: Baseline forest 2017

Map 2 Raw Hansen loss 2024

Map 3: Clean loss 2024 (≥ 0.5 ha)

Conclusions – Leakage Assessment (2017–2024)



Sources

NDVI classification:

- Tucker 1979
- Pettorelli 2013
- USGS NDVI
- Forest et al., 2018; Jiang et al., 2006
- Gao 1996; Sánchez-Ruiz et al., 2020
- (supported by standard exploratory data analysis practice, e.g. Tukey, 1977).